

ARTICLE

3D-printed PLA/Gel hybrid in liver tissue engineering: Effects of architecture on biological functions

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Abstract

The liver is one of the vital organs in the body, and the gold standard of treatment for liver function impairment is liver transplantation, which poses many challenges. The specific three-dimensional (3D) structure of liver, which significantly impacts the growth and function of its cells, has made biofabrication with the 3D printing of scaffolds suitable for this approach. In this study, to investigate the effect of scaffold geometry on the performance of HepG2 cells, poly-lactic acid (PLA) polymer was used as the input of the fused deposition modeling (FDM) 3D-printing machine. Samples with simple square and bioinspired hexagonal cross-sectional designs were printed. One percent and 2% of gelatin coating were applied to the 3D printed PLA to improve the wettability and surface properties of the scaffold. Scanning electron microscopy pictures were used to analyze the structural properties of PLA-Gel hybrid scaffolds, energy dispersive spectroscopy to investigate the presence of gelatin, water contact angle measurement for wettability, and weight loss for degradation. In vitro tests were performed by culturing HepG2 cells on the scaffold to evaluate the cell adhesion, viability, cytotoxicity, and specific liver functions. Then, high-precision scaffolds were printed and the presence of gelatin was detected. Also, the effect of geometry on cell function was confirmed in viability, adhesion, and functional tests. The albumin and urea production of the Hexagonal PLA scaffold was about 1.22 ± 0.02 -fold higher than the square design in 3 days. This study will hopefully advance our understanding of liver tissue engineering toward a promising perspective for liver regeneration.

KEYWORDS

3D printing, bioinspired, gelatin, liver cell function, liver tissue engineering, PLA

1 | INTRODUCTION

The liver is the largest organ in the human body that performs various functions such as protein production and metabolic processes playing a significant role in human health and longevity. However, its ability to regenerate chronic/acute diseases or viral infections may cause permanent loss of liver function (Yoon No et al., 2015). Chronic liver disease affects 500 million people yearly and yields 2% of worldwide

deaths (Aleahmad et al., 2017; Mazza et al., 2017). A severe shortage of organs for transplantation and the encountered complications have increased the medical need for tissue engineering organs. This deficiency grows in all body parts, especially the liver (Hoganson et al., 2010; Kryou et al., 2019; Zakeri et al., 2020).

One of the promising approaches of tissue engineering to solve the challenge of liver failure is using artificial scaffolds as a substrate for the growth and proliferation of liver cells. The term "liver tissue

engineering" or "LTE" sums up one of the ultimate goals of modern biotechnology: the possibility of complete or partial reproduction of liver function for the treatment of acute or chronic liver disorders and, ultimately, the creation of a fully functional organ for transplantation or an ex vivo bioartificial device. All technical approaches in liver tissue engineering are based on allocating adult hepatocytes or hepatocyte-like cells from stem cells in a three-dimensional structure that can ensure their survival and maintain their functional phenotype (Mazza et al., 2017). However, the complications of using these cell sources have led researchers to use cell lines such as HepG2 for primary evaluation of their systems (Lewis et al., 2018; Ma et al., 2016; Mirdamadi et al., 2020; Tahmasbi Rad et al., 2014). Progress in the field of LTE has brought about much valuable information regarding in vitro and in vivo growth as well as liver biology, pathology, and cell-biomaterial interactions (da Silva Morais et al., 2020; Lewis et al., 2018; Moradi et al., 2020).

Bioinspired strategies have revolutionized tissue engineering with their inherent high potential (Eftekhar et al., 2020; Tavakol et al., 2016). The essential aspects of liver scaffold characteristics include mimicking the chemistry of extracellular matrix components and the specific physical interactions between the scaffold and cell surface receptors. Therefore, the design and selection of materials for liver scaffolds have gained more attention in recent studies (Kang et al., 2018; Lewis et al., 2018; Mirdamadi et al., 2020; Rose et al., 2021; Xia et al., 2018; J. Zhang, Zhao, et al., 2018). To obtain higher levels of liver function and mechanical stability, cultured hepatocyte cells on various types of biological substrates such as hydrogels (Jeon et al., 2017), porous scaffolds (German & Madhally, 2018), microcapsules (Zhu et al., 2017), and hollow fibers (Ahmed et al., 2017) have been studied.

Poly-lactic acid (PLA), an FDA-approved polymer, has been chosen as one of the best materials for designing biomaterials due to its biocompatibility and biodegradation (Ishii et al., 2008). Among biodegradable thermoplastic polymers for fused deposition modeling (FDM), PLA is mostly used for soft tissue engineering because of its higher hydrophilicity compared to PCL, hence the surface properties for cell adherence are superior (Patricio et al., 2013). However, its poor surface properties have led scientists to incorporate different coatings (Mozafari et al., 2015). In a study to improve the surface and mechanical properties of the poly-L-lactic acid (PLLA) porous scaffold, a coating process was performed after alkaline hydrolysis of the surface (Guo et al., 2013). Among the various surface modification methods, physical entrapment is studied for loading bioactive macromolecules on the surface of polymers without needing active sites. In this way, gelatin molecules, which can improve adhesion and growth, were coated by Zhang et al. on micropatterned poly(L-lactide-co-ε-caprolactone) (PLCL) film. Under these conditions, high-yield and stable gelatin molecules were placed on the film and trapped (D. Zhang, Xu, et al., 2018).

The introduction of 3D printing, or a more general term additive manufacturing (AM), has empowered tissue engineering to develop living systems with precision on multiple scales (Asadi-Eydivand et al., 2016; Baltazar et al., 2020; Bashiri et al., 2021; Matai et al., 2020). Among various AM techniques, FDM is a well-known

and most common biofabrication method, which is a layer-by-layer extrusion-deposition process for thermoplastic materials with steady flow (Ramos & Moroni, 2020; Seyedsalehi et al., 2020; Zare et al., 2022).

Deployment of the AM technique allows tailoring the scaffold's structure in a bioinspired way. This is particularly important in the liver tissue, and the effect of the structure has been studied on the cellular response and shown that pseudo-hexagonal geometries positively influence hepatocyte functions (Lewis et al., 2018; Ma et al., 2016).

Hepatocyte polarization and enzymatic activities of a 3D bioplotting PLLA-Collagen scaffold with a 30° strut angle between subsequent layers were superior compared to a sandwich system (B. Wang et al., 2016). Lewis et al. have used a hexagonal-shaped 3D-printed gelatin scaffold for liver tissue engineering. They showed that it expressed better hepatic functions like gene and albumin expression, bile canaliculi formation, and so forth, compared to a simple-designed scaffold (Lewis et al., 2018). Using photo-pattern technology, Tsang et al. produced a multilayer hexagonal structure to enhance oxygen and nutrition transportation (Tsang et al., 2006). Also, gradient channel structures mimicking the branched vascular network using different cells for a specific function could reach a functional endothelial network in vitro and in vivo compared to a non-gradient structure (Zhu et al., 2017). Moreover, the effect of architecture and design has been proven in various studies (Mirdamadi et al., 2020).

This study investigates the function of the HepG2 liver cell line in 3D-printed PLA scaffold systems printed by the FDM with gelatin coating. The aim was to establish a relationship between scaffold architecture and the properties raised from it as well as cellular functions. The scaffolds have been designed to test the project hypotheses considering the liver lobule's hexagonal shape. The motive of using a common technique with low complexity as a smart system for culturing liver cells has been a guideline for this study. For this purpose, physical, chemical, and biological evaluations have been conducted, which are thoroughly studied here. Figure 1 represents the summary of design for this project and its results.

2 | EXPERIMENTAL

2.1 | Materials

In this research, PLA filament (with a density of 2.6 g/cm³ and a molecular weight of 60,000 g/mol) and glutaraldehyde from Sigma-Aldrich, gelatin powder, acetone, and ethanol from Merck, and sodium hydroxide from SCP were purchased.

2.2 | 3D design of the scaffold

The scaffolds were designed by Solidworks software. There are two main designs for this project. The first design has a square (S), and the second design with a hexagonal (H) cross section. The dimensions of the H design have been adjusted to obtain the smallest hexagon

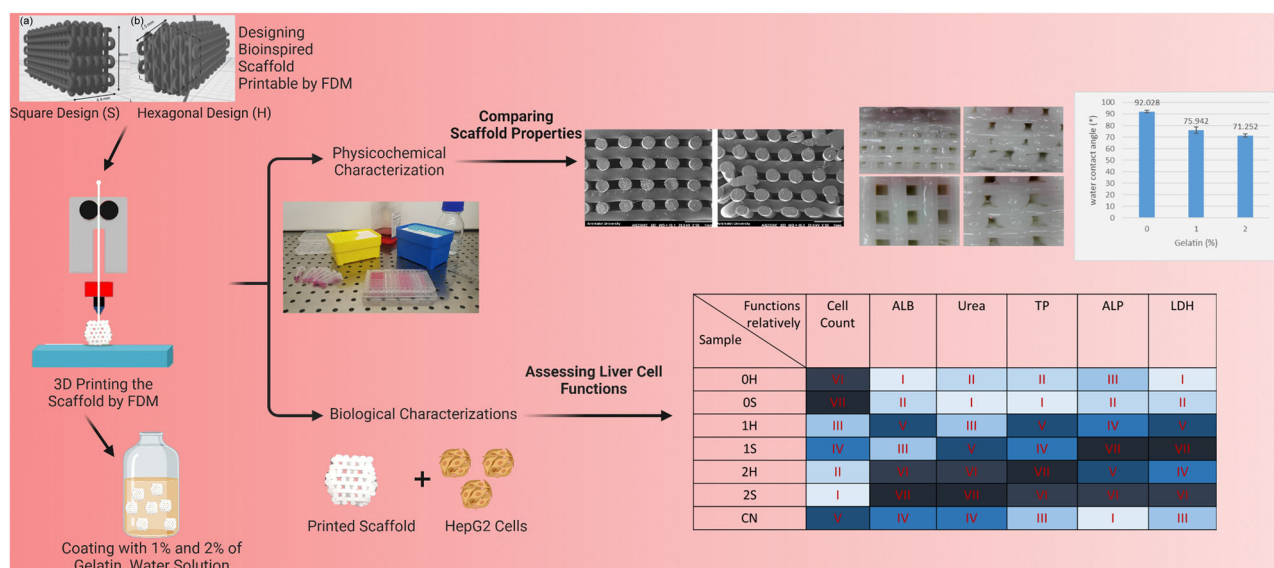


FIGURE 1 Summary of design, from left to right: the fabrication process including the design, three-dimensional printing, and gelatin coating; characterization divided to physicochemical (left to right: scanning electron microscopy images, microscopic images, and water contact angle in deferring gelatin coating scaffolds), and biological (dark blue to light blue indicated higher to lower content of the function presented in the column, comparing hexagonal [H] and square [S] design with different gelatin content [0%, 1%, and 2%]) to investigate the scaffold properties and respective function for liver cells. The roman numbers are for more clarity to distinguish between the colors.

Design	Volume (mm ³)	Surface area (mm ²)	Surface area/volume (mm ⁻¹)	Strut diameter (mm)	Space between struts (mm)
S (Square)	141.61	1418.83	10.02	0.4	0.4
H (Hexagonal)	135.94	1440.65	10.59	0.4	0.4

TABLE 1 Dimensions of the scaffolds

possible, and the dimensions of the S design have been chosen to equalize the surface-to-volume ratio of the scaffolds with two different designs. Table 1 lists the values given for the software for the two designs. After designing the scaffolds, the files were saved in STL format for later use.

2.3 | 3D printing of the scaffold

The STL file from Solidworks software was converted into a G-code file by Slic3r software (Figure 2). The designed scaffolds were printed using the PLA filament by the FDM printer (Quantum 3D-Generous) with the nozzle and a bed temperature of 195 °C and 50°C, respectively, and a nozzle diameter of 0.4 mm.

2.4 | PLA/Gel hybridization entrapment (M-Entrapment)

According to the protocol introduced by Cui et al. (2003), briefly, in the entrapment method, 7 to 3 volumes of acetone and distilled

water were prepared. The scaffolds were then immersed in this solution for 1 h. They were immediately incubated in an aqueous gelatin solution of 1% and 2% for 24 h at 37°C. Then this method was modified to the M-Entrapment method by placing the samples in a -80°C freezer and freeze-drying for 4 h. Then they were crosslinked by immersion in a 1% water solution of Glutaraldehyde for 2 h, followed by washing with deionized water. Afterward, the scaffolds were sterilized using UV light on each side for 20 min.

2.5 | Characterization of the PLA/Gel hybrid

To investigate the geometrical structure of the scaffolds, their surface and cross sections were first observed by a light microscope (Optix-Z5). Then scanning electron microscopy (SEM-Seron Technology AIS-2100) was used to examine the scaffolds with and without gelatin with both designs. To achieve a smooth and flat surface for better energy dispersive spectroscopy (EDS) and contact angle measurements, PLA films were prepared, and hybridization was performed, as mentioned earlier.

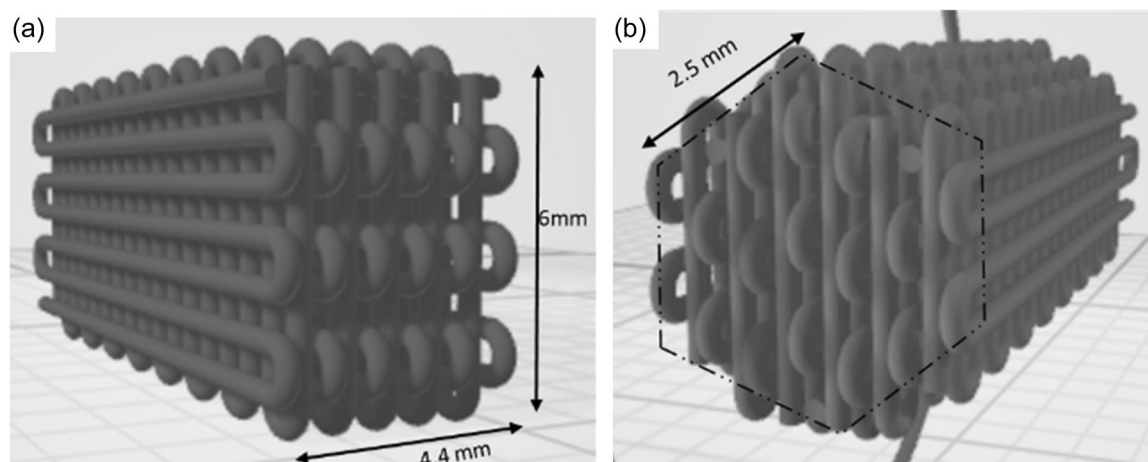


FIGURE 2 Designs of scaffolds and their dimensions (a) square and (b) hexagonal cross-sectional scaffolds in Slic3r

The composition of the materials used and the presence of gelatin was confirmed by the EDS (SAMX) test for PLA, PLA-1% Gel, and PLA-2% Gel samples before and after crosslinking for determining the necessity of crosslinking the gelatin. Water contact angle measurement determined the surface hydrophilicity of the material. In all, 0.005 ml of water was poured onto the specimens, and then after 7 s, images were taken from the droplet, and the contact angle of the specimens was measured by ImageJ software. This was performed 10 times for each sample, and the results are the average of the repetitions.

2.6 | Cell culture

HepG2 cell line was cultured in Dulbecco's modified Eagle's medium with high glucose containing 10% fetal bovine serum and 0.5% Pen/Strep in flasks. The medium was then emptied, and after washing with phosphate-buffered saline (PBS), 0.25% Trypsin was added for cell separation and counted after centrifugation. Then 8000 cells were plated on sterilized scaffolds in 48-well plates, and after 1 h incubation, 100 μ l of complete medium was added to them and incubated at 37°C with 90% humidity and 5% CO₂. For the cellular evaluations, the samples were coded according to Table 2.

2.7 | MTT assay

MTT test was performed on specimens with 0%–1%–2% gelatin with both designs and a polystyrene plate as a control sample. At this stage, 4000 cells were plated on each scaffold in a 96-well plate, and after 1 and 8 days, 100 μ l of MTT solution was added to each cell, then DMSO was added after 4 h of incubation. After 15 min, the purple formazan precipitate was read by the ELISA Plate Reader. The test also averaged three replicates per interval for each sample (Simorgh et al., 2021).

TABLE 2 Experimental group coding

Code	Group description
0H	No gelatin hexagonal scaffold
0S	No gelatin square scaffold
1H	1% Gelatin hexagonal scaffold
1S	1% Gelatin square scaffold
2H	2% Gelatin hexagonal scaffold
2S	2% Gelatin square scaffold

2.8 | Cell count

HepG2 cell adhesion was measured by counting the number of adherent cells. At this stage, in the first, third, and seventh day after cell culture, the cells on scaffold samples with two designs and 0%–1%–2% gelatin were counted, and the polystyrene plate was the control sample. The culture media was washed with PBS, stained with trypsin, and then counted after adding medium to neutralize the trypsin. For this test, three replicates were prepared from each sample, and their mean was reported to give more accurate results.

2.9 | Characterization of liver functions

To evaluate the function of liver cells, the secretion of liver biomolecules, including ALB (Albumin), urea, LDH (lactate dehydrogenase), TP (total protein), ALP (alkaline phosphatase), AST (aspartate aminotransferase), and ALT (alanine aminotransferase) were measured. The levels of each of these biomolecules were measured in the laboratory of the Shohada Tajrish Hospital by DiaSYS German kits. In the first, third, and seventh day after cell culture, the supernatants of all samples were collected, and the biomolecules were measured. The polystyrene plate was used as the control sample.

2.10 | Accelerated degradation

To investigate the degradation, 0.25 mM NaOH solution at pH 12.68 was prepared according to the research done on PLA. First, all samples were weighed, and 50 ml NaOH solution was added per gram of the scaffolds (Tisserat & Finkenstadt, 2011). Then the weight of each sample and pH of the medium was measured at 2, 24 (1 day), 72 (3 days), 168 (7 days), and 336 (14 days) hours. The degradation rate (weight loss percentage) was calculated according to the following formula (Equation 1):

$$m_2 - m_1 / m_1 \times 100. \quad (1)$$

In this formula, m_1 is the initial weight, and m_2 is the secondary weight of the scaffold after being placed in the degradation solution and drying for 24 h. In this test, the medium was changed weekly, with three separate replicates for each scaffold, and the reported data were averaged.

2.11 | Statistical analysis

All experimental data are presented as mean $\pm$ SD. Statistical analysis was performed using a one-way analysis of variance (one-way ANOVA) using the Tukey HSD post hoc test and homogeneity by Duncan test. A value of $p < 0.001$ was considered significantly different. Tests were conducted using SPSS 16.0 software (SPSS).

3 | RESULTS AND DISCUSSION

3.1 | PLA 3D-printed scaffolds

As shown in Figure 3, the printed scaffolds were similar to the design intended, and the dimension accuracy was shown by both the SEM images (Figure 4) and optical images (Figure 3).

As shown in Figure 4a,c, the arrangement of the struts in a square design in each layer is precisely on the top of the other layer's strut, which allows pores to be continued in a linear pattern. In contrast, in the hexagonal design (Figure 4b,d), the struts of each layer are not precisely on the top of the next layer's strut but on the second layer's strut, giving rise to continued porosities with a zigzag pattern in the z-axis. Moreover, in Figure 4, in the hexagonal design, in addition to the overall shape of the scaffold's section, the struts' pattern also follows the hexagonal shape. Likewise, in the square design, the struts are square. The main difference between our design and previous studies (Lewis et al., 2018) is that the square and hexagonal are shaped in the cross-section of the scaffold. It is noteworthy that these images (Figure 4c,d) also show the design conformity (Figure 4a,b) by the printed scaffolds due to the precise printing process according to the machine and software settings.

3.2 | Characterization of the hybrid scaffolds

Previously the effect of different coating methods on the properties and function of the 3D printed scaffold has been studied (Mirdamadi

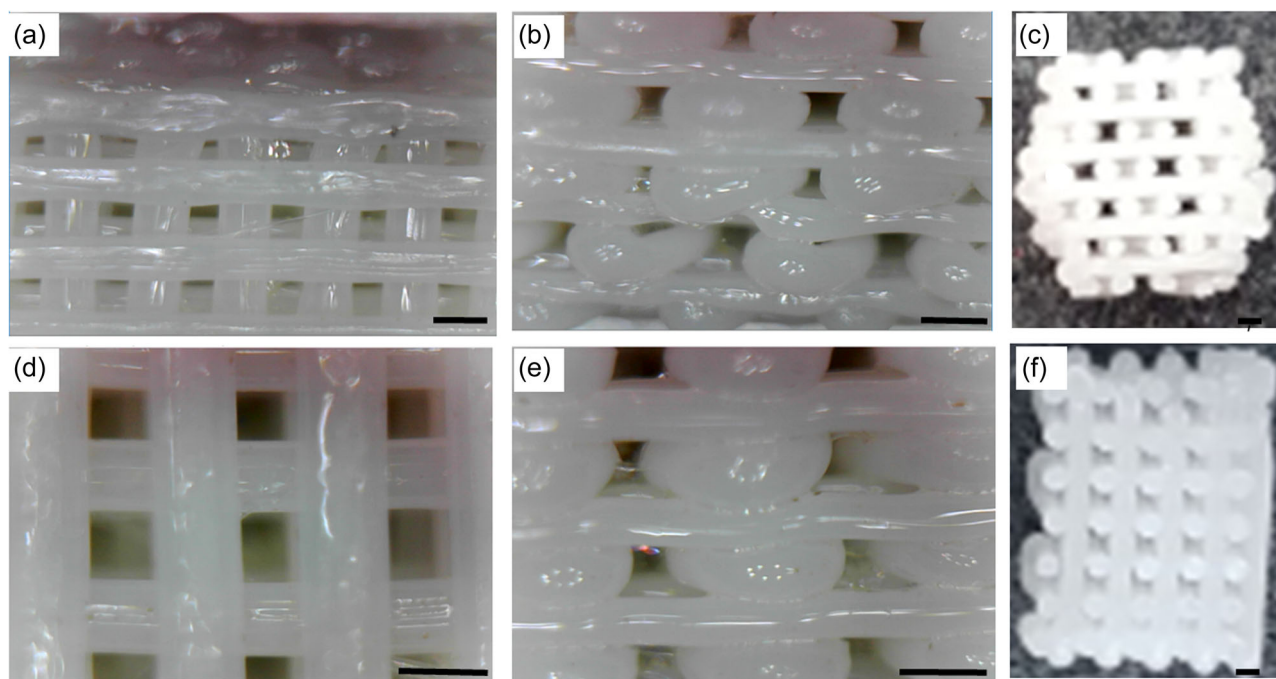


FIGURE 3 Images of the printed scaffolds: (a–c) hexagonal and (d–f) square designs. (a, d) The top, (b, e) side, and (c, f) cross-sectional area of the scaffolds. Black lines are 0.4 mm.

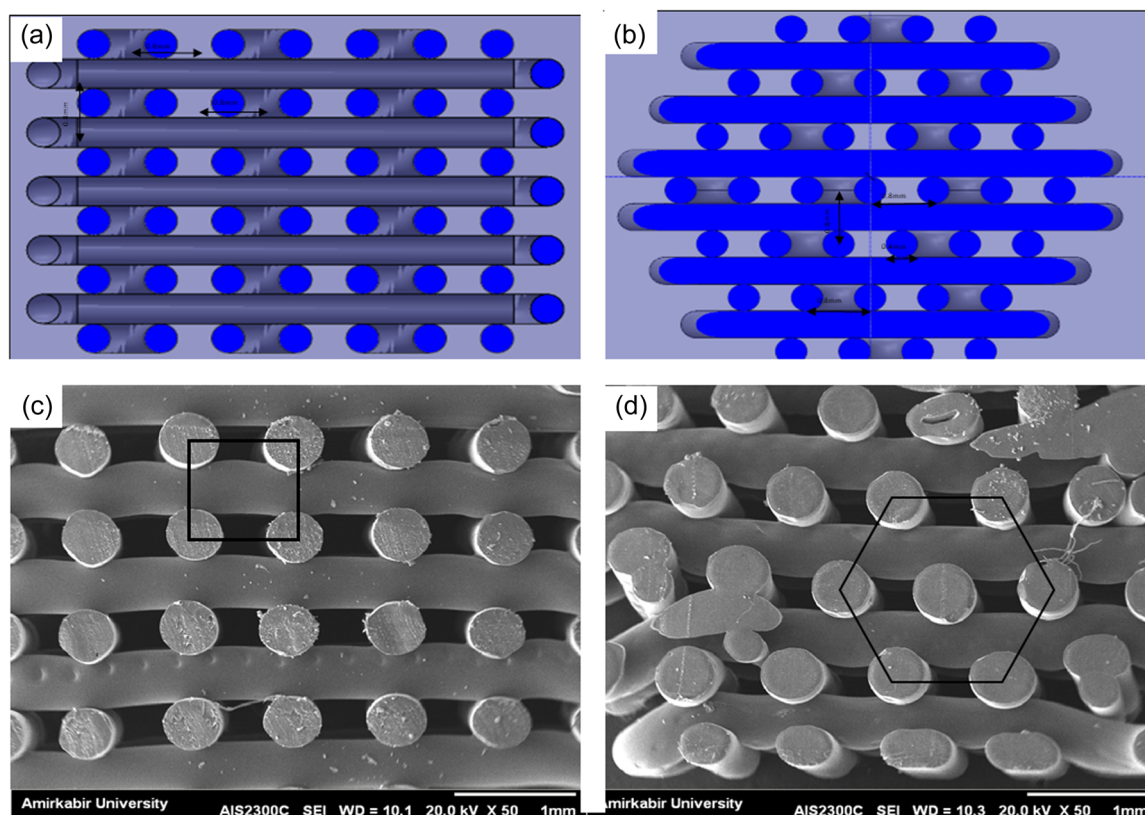


FIGURE 4 Cross-sectional view of (a, c) square and (b, d) hexagonal design scaffolds. (a, b) Solid works design and (c, d) scanning electron microscopy images, all of them with 0.8 mm space between the center of the struts with the same design and 0.4 mm diameter struts.

et al., 2022). For a closer examination of the effect of gelatin coating on the physical properties of the surface and topology images have been taken before and after the coating and after exposure to culture medium. Figure 5a,b shows that in the Entrapment method, due to the initial hydrolysis, some closed pores are created on the surface, increasing the surface roughness which is also reflected as an opaqueness comparing Figure 5f and Figure 5e taken by light microscopy. The size of the porosity on the hydrolyzed surface increased after 7 days, indicating the surface degradation of the scaffold. Also, the hydrolysis process of Entrapment method has resulted in reducing the diameter of the struts and started degrading the scaffold shown in Figure 5c,d from the surface. Furthermore, contact angle test for the evaluation of hydrophilicity has been conducted. The contact angle measurement of the liquid droplets on the surface is used to characterize the surface's wettability, cleanliness, and hydrophobicity/hydrophobicity. The contact angle depends on the sample's surface energy and the tensile stress of the drop. The higher the wettability level, the closer the contact angle is to zero; hence, the droplet spreads on the surface. Composite materials may not have a homogeneous surface, so contact angle measurements are performed at 10 points and reported as the average (Li et al., 2004). Figure 6 shows the mean contact angle of each sample. With increasing the gelatin content, the water contact angle on the prepared film decreases, indicating an increase in surface hydrophilicity. As we know, PLA has low surface energy

hydrophilicity and lacks detectable signals for cells. One way to modify its surface is by coating it with hydrophilic materials such as gelatin or surface hydrolysis (Cui et al., 2003; Guo et al., 2013). As a result, we expect an increase in the contact angle in the gelatin-coated specimens.

The EDS spectrum of the samples prepared with the weight and atomic percentages of N, C, and O elements are further presented to confirm the presence of gelatin in the structure (Table 3). Given the chemical structure of PLA and gelatin, we know that the only difference between the elements in the chemical formula of these two substances is the presence of the N in gelatin which is not present in PLA (Mahapatro & Singh, 2011; Stoessel et al., 2015). Therefore, this test compared the percentage of N in these samples. Due to the presence of N in the environment, the device detects it in all samples. However, its increase is an indication of gelatin.

According to the results of the EDS test, in all samples containing gelatin except 2% gelatin before crosslinking, atomic percentage—At % and weight percentage—Wt% of N were higher than in the PLA sample. In 1% gelatin, crosslinking reduced the amount of gelatin present on the surface, most likely due to the solubility of gelatin in the water solution of glutaraldehyde. This suggests that due to the presence of 99% water in the glutaraldehyde solution, some of the gelatin on the surface was washed with water before crosslinking. Also, the N content of samples with 2% gelatin before crosslinking was not changed, indicating that gelatin will not remain on the

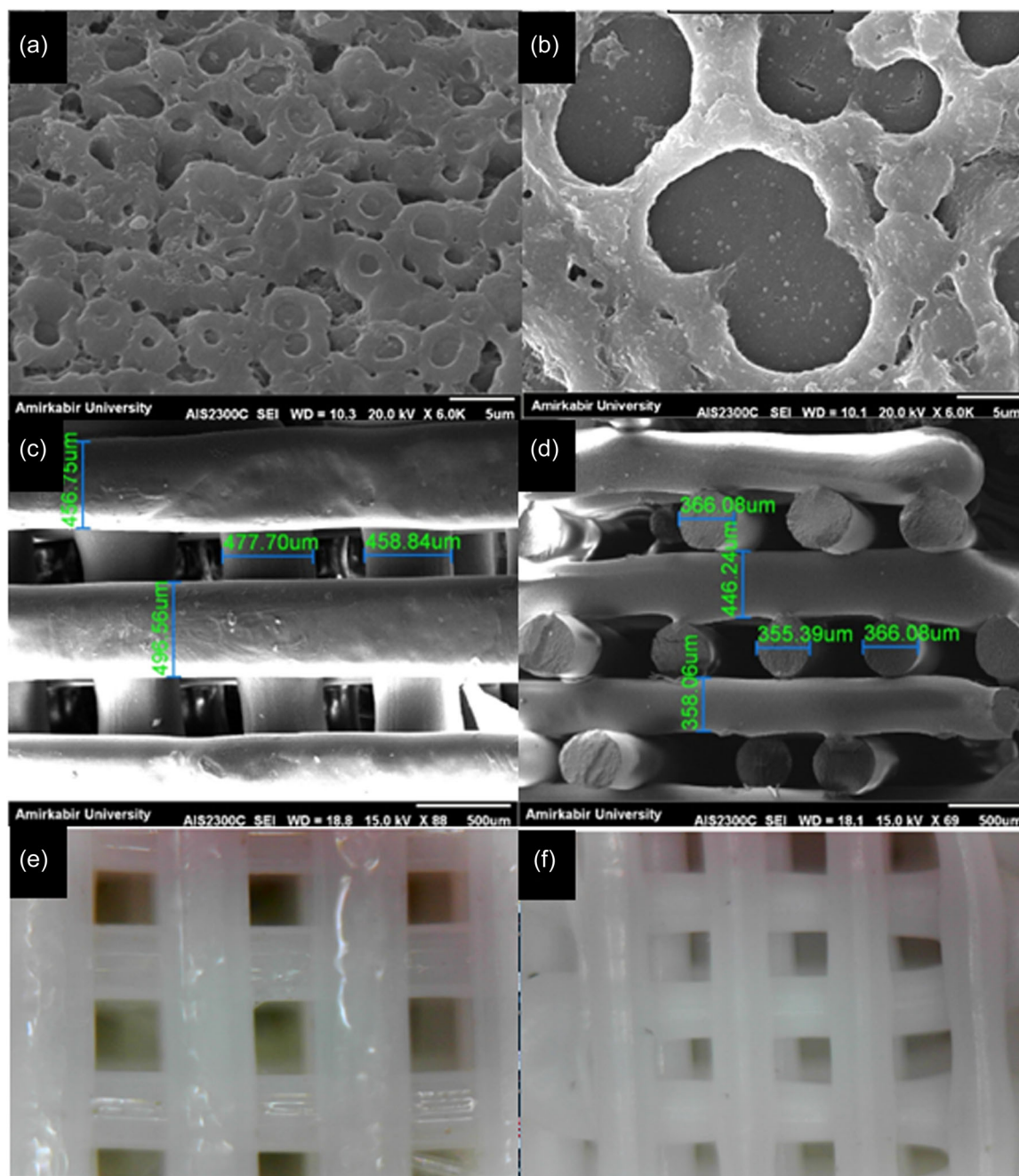


FIGURE 5 The surface of the scaffolds coated with gelatin by M-Entrapment after (a) 1 day and (b) 7 days. Scanning electron microscopy images of a scaffold (c) before and (d) after surface hydrolysis with acetone/water. Light microscopy images of the scaffold's surface (e) before and (f) after gelatin coating.

surface without crosslinking due to its higher concentration and thus higher weight and density. As expected, the N percentage in the scaffold with 2% gelatin coating after crosslinking was the highest.

3.3 | Adhesion and proliferation

The following SEM images (Figure 7) and diagram (Figure 8) show the extent of cell adhesion by counting the number of cells in different

samples and days. As shown in Figure 8, all the samples except the scaffolds containing 2% gelatin showed ascending adhesion over time. Accordingly, the number of cells on Day 7 in gelatin-free samples was higher than in all samples. This increase in adhesion can be justified by the tendency of hepatocytes to aggregate (Lewis et al., 2018). In other words, in the presence of gelatin, the cells adhere to the surface and become isolated on a single scaffold. Due to the absence of adjacent cells and the signals derived from this affinity, their activity and proliferation reduce. In contrast, in the

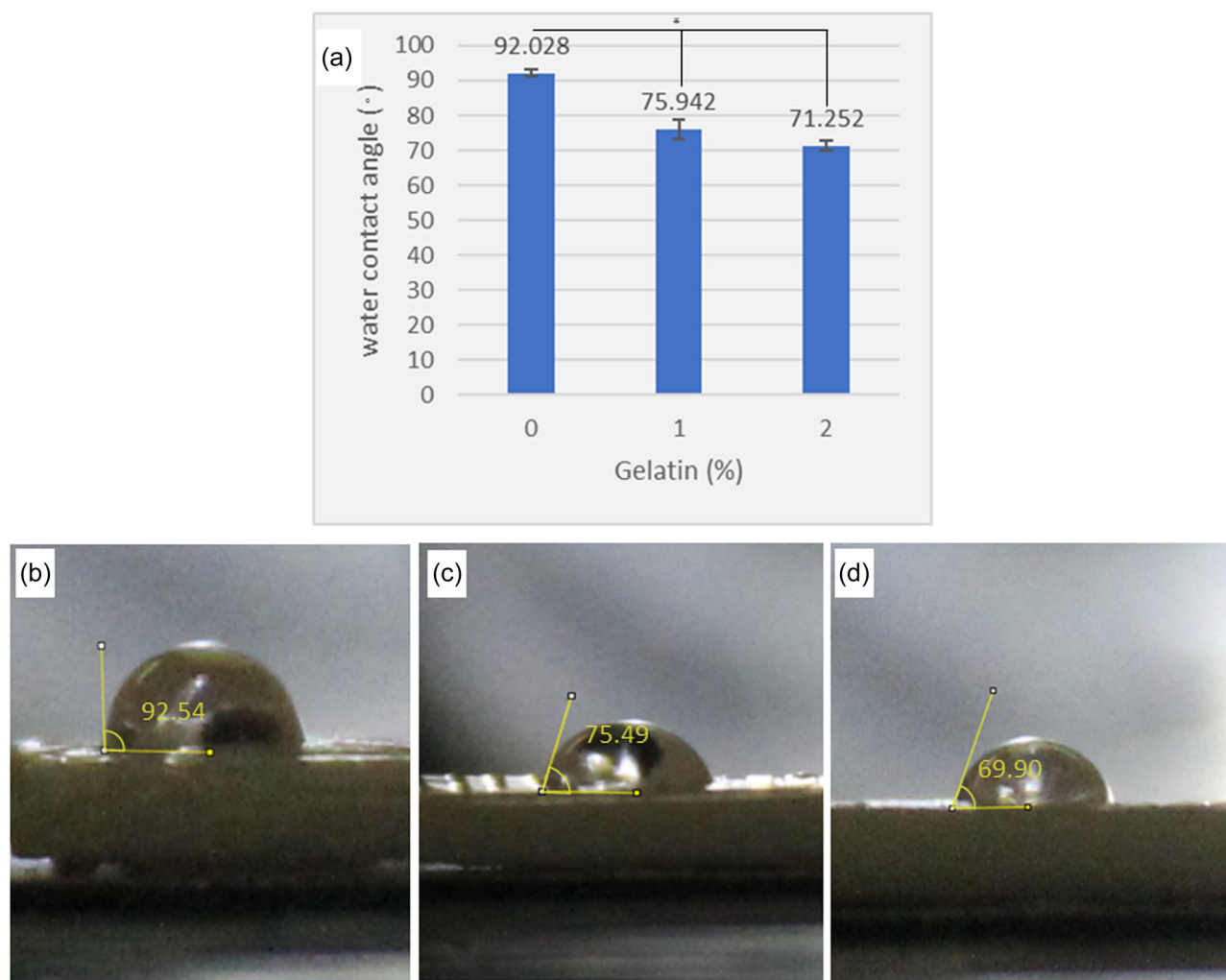


FIGURE 6 (a) Water contact angle of poly-lactic acid films with 0%, 1%, and 2% gelatin coating. Images of water droplets on samples with (b) 0%, (c) 1%, and (d) 2% gelatin coating. * shows p value < 0.05. Sample number $n = 5$.

TABLE 3 Weight (Wt%) and atomic (A%) percentage of Nitrogen element by EDS

Samples	PLA	Before crosslinking		After crosslinking	
		PLA-1% Gel	PLA-2% Gel	PLA-1% Gel	PLA-2% Gel
A% (N)	15.22	29.25	13.18	20.58	33.98
Wt% (N)	15.29	30.31	13.41	20.98	34.92

absence of gelatin, the cells aggregate together and increase cell signaling, increasing their proliferation. The highest adhesion was observed for OS and 2S on the third day and OS and OH on the seventh day.

The trend of gelatin-free samples' cell adhesion increases over time, which means an increase in proliferation these days. In addition, the square scaffold was more adherent than the hexagonal scaffold and was better than the control on the third and seventh days, with significant differences. The design and geometry of the scaffold at this scale in the gelatin-free samples affect cell adhesion. Contrary to the initial assumption, the square cross section has a better adhesion performance in gelatin-free samples. This may be due to the initial

efficiency of cell seeding in the square section sample as the gravitational forces are more prone to affect cells to lay on the horizontal strut of the scaffold with square cross section (with an angle of 90°) rather than the steep strut of the hexagonal cross-sectional scaffold (with an angle of 60°). These data are in line with the previous study assessing the seeding efficiency in scaffolds with different strut angles (Lewis et al., 2018).

Samples containing 1% gelatin showed less cell adhesion than the control sample, and the 1H sample had more cells attached. However, like the gelatin-free, the square cross-sectional scaffold on Day 7 had better adhesion results and similarly demonstrates the role of geometry on cell function.

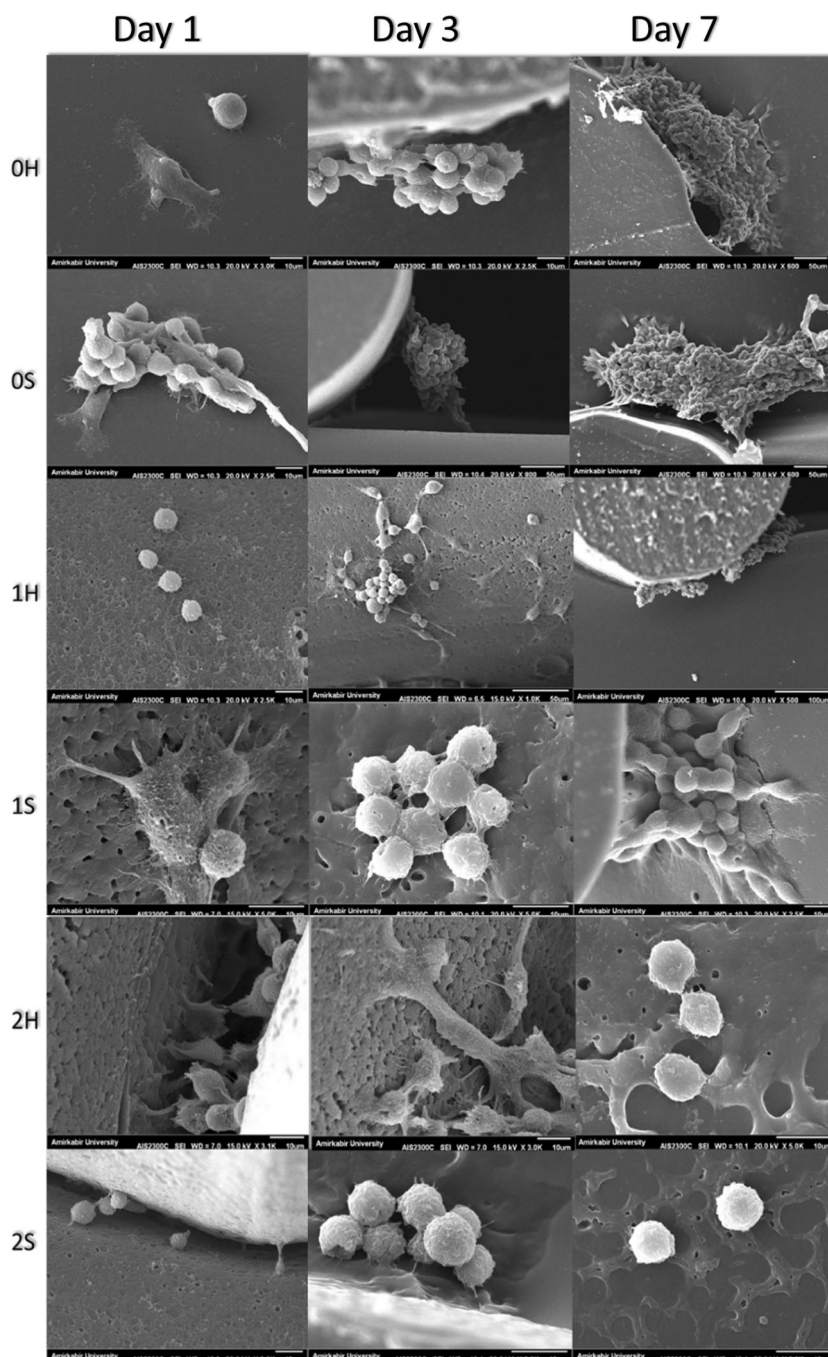


FIGURE 7 Scanning electron microscopy images of the scaffold after culturing HepG2 cells for Square (S) and Hexagonal (H) samples with 0%, 1%, and 2% gelatin coating after 1, 3, and 7 days of culture

In the case of scaffolds containing 2% gelatin, the adhesion shows that cells were initially attached to the scaffold and could grow and reach the control sample. Their numbers had declined even below the control sample by the seventh day. This can be due to the destruction of gelatin from Day 3 onwards, which results in the cells adhering to it being removed and the number of cells decreased. Gelatin is the only water-soluble component in the scaffold and as the culture medium is a water solution it is concluded that the gelatin is degraded over time. Studies have shown that gelatin crosslinked with GA (glutaraldehyde) degrades after about 7 days of incubation in PBS (Nouri-Felekori et al., 2019). This degradation is also supported by the SEM images in Figure 6a,b.

3.4 | Cell viability and cytotoxicity

Based on the level of absorption determined by ELISA Plate Reader for each scaffold, viability% was obtained by MTT assay as shown in the following diagram (Figure 9) ($p < 0.05$). According to this chart, all specimens had a viability of over 100%, which means they showed no toxicity. Among these samples, the highest survival rate was for the 1H sample on the first and seventh days.

By comparing all samples with the same gelatin percentages, it can be observed that the viability was higher in all hexagonal scaffolds except for 0S. Also, the viability has diminished over time. These findings also show the effect of geometry on scaffolds

FIGURE 8 Numbers of cells attached to the samples counted in different days showing the adhesion and proliferation of HepG2 cells. In Day 1 the control sample had a significant difference ($p < 0.05$) with 0H, 1H, 2H, and 2S. In Day 7 0H had significant difference ($p < 0.05$) with all other samples except 0S. In Day 7 0S had significant difference ($p < 0.05$) with all other samples except 0H (sample number $n = 3$).

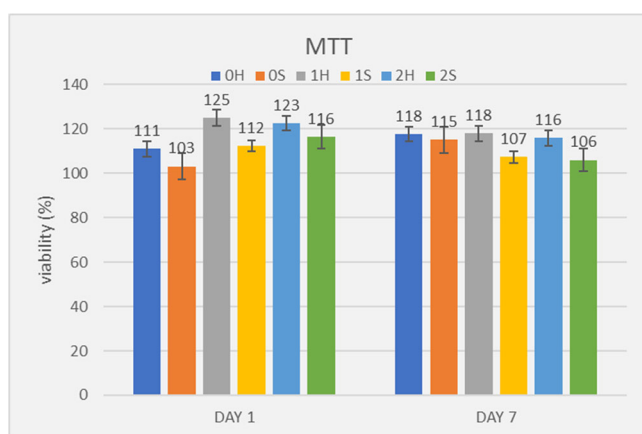
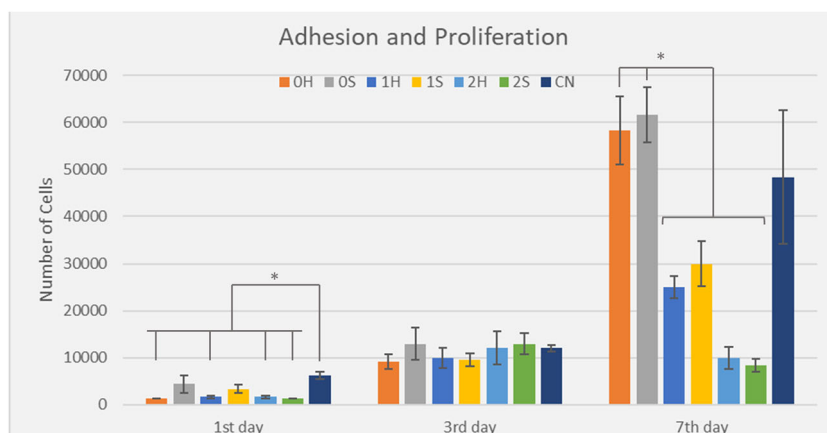


FIGURE 9 Viability% of all samples in Days 1 and 7 was assessed by MTT. No significant difference between samples in a given day. Sample number $n = 3$.

function and, unlike the adhesion, scaffolds with hexagonal cross sections had higher cell viability. Cell viability increased over time in square gelatin-free scaffolds and decreased in scaffolds containing 1% and 2% gelatin. This decrease in survival in gelatin-containing samples may be due to gelatin degradation during these 7 days.

Regarding the survival rate in the hexagonal cross-sectional samples (Figure 9), no significant changes were observed with the gelatin percentage change. The viability in the 1H sample was the highest the whole time showing the effectiveness of scaffold design in supporting cell survival. However, further assessment is needed to determine the efficiency of this system in supporting cell survival and function. Also, although the 2S sample had the most increased survival at the beginning, the 1S sample was better than the other two S samples on the seventh day. Hence, adding gelatin has more effect on cell viability in the square than in hexagonal scaffolds.

Cytotoxicity indicates damage caused to the cell membrane; hence, necrosis can be determined by LDH secretion, a liver damage marker, which acts as a soluble cytosolic enzyme in the culture medium (Kryou et al., 2019; Y. Zhang et al., 2016). Also, AST and ALT

are known as hepatotoxic markers. Comparing the release of these markers in different samples can show the level of cytotoxicity in the sense of cell damage in HepG2 cells (Kermanizadeh et al., 2017; Kitchin et al., 2020). The damage caused by differing designs was not studied before in this way; however, cell viability in different cell printings with various architectures has demonstrated that the stress caused by sharp edges can lessen the viability (Agarwal et al., 2021). Thus, the increased amount of LDH, AST, ALT, and ALP for the Hexagonal design samples can be justified this way since the chemical properties of samples with the same percentage of gelatin are the same.

As shown in Figure 10, the LDH level of samples in the first three days was not altered much, but on the seventh day, the gelatin-free samples had a relatively higher level of LDH secretion. These results comply with the AST and ALT protein concentration on the seventh day. These data, along with cell adhesion, suggest that though there were a high number of cells on 0H and 0S samples, the damage inflicted on cells was also increased. Hence, the PLA scaffolds supported the proliferation of cells, but their proper function was diminished over time, later accompanied by the liver cell function results. Combining the damage caused by design with the support of liver function in gelatin-coated samples indicate that the hydrogel coating can enhance cell function and give the bioinspired design a chance to promote specific liver function.

ALP acts as a phosphate monoester hydrolase in the body to remove phosphate groups and catalyze the hydrolysis of phosphate esters in alkaline conditions. The liver, intestine, bone, and placenta are the main organs in which ALP is found (Dasgupta & Wahed, 2013; Y. Wang et al., 2020). ALP elevation in the body is considered a signal of liver or bone disorders. In HepG2 cells, ALP has been assessed to determine the effectiveness of a drug for treating these cancer cells (K. Wang et al., 2018). In Figure 10, ALP production for sample 1S was the highest on Day 7 and has the most elevation. Hexagonal scaffold showed higher ALP secretion only on Day 7 for gelatin-free scaffolds indicating that the secretion of this biomarker is more dependent on the chemical properties of the scaffold in this system.

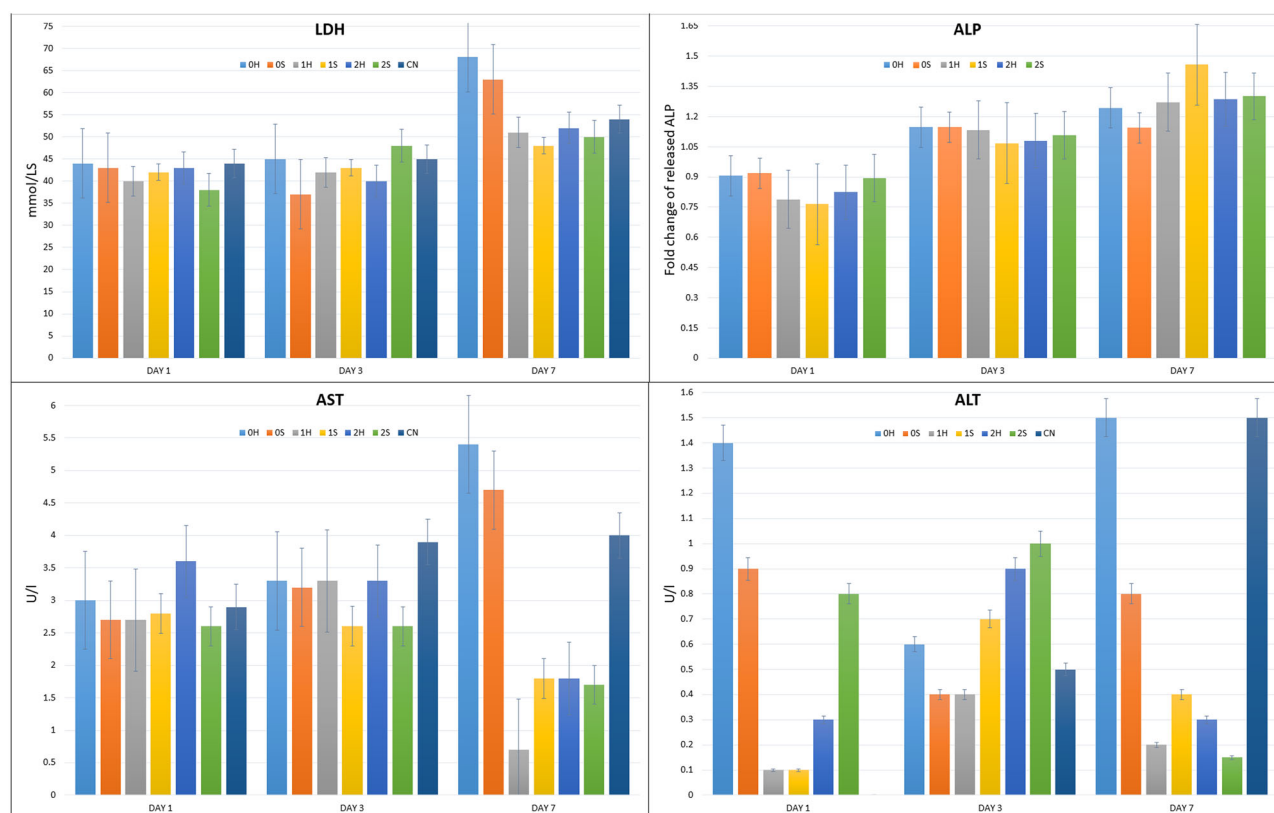


FIGURE 10 Cytotoxicity assessed by LDH concentration in all samples on Days 1, 3, and 7. Hepatotoxicity was assessed by AST, ALT, and ALP secretion on Days 1, 3, and 7 for all samples. Sample number $n = 3$. ALP, alkaline phosphatase; ALT, alanine aminotransferase; AST, aspartate aminotransferase; LDH, lactate dehydrogenase.

3.5 | Liver cell functions

One of the liver's essential functions is the secretion of biomolecules such as albumin, urea, TP, and other proteins. These biomolecules are secreted by the hepatocytes with the help of nonparenchymal cells and are recognizable in vitro (Usta et al., 2015). The diagrams in Figure 11 show the secretion of each of these substances in different samples relative to cell count on the first, third, and seventh day of culture. The activity of these biomolecules in this study defines the liver function of HepG2 cells.

As the most redundant protein in the plasma, human serum albumin comprises 50% of the whole proteins. The liver is specialized in synthesizing and releasing albumin directly into the blood without storing it. With a tremendous function, 9–12 g of albumin is secreted by 20%–30% of hepatocytes a day which can be increased by 3–4 times if necessary (Bernardi et al., 2012).

As seen in all the diagrams in Figure 11, the secretion of these biomolecules is high on the first day and decreases over time. However, except for the seventh day, all samples had more functions than the control group. Notably, 2% gelatin samples on Day 7 had the most concentration of biomarkers despite low cell count (Figure 8), suggesting that liver cell function has been preserved more efficiently in these samples. The same results in other studies showed that stiffer surfaces resulted in functionality

loss over time (Hwang et al., 2021), which can explain our research.

According to the microalbumin secretion diagram (Figure 11—ALB), it can be observed that all the samples have shown more albumin secretion per cell than the control group except on the seventh day, and they are all descending over time. On Day 1, 0H and 1S and on Day 7, 2S and 2H had higher albumin secretion per cell. The hexagonal scaffold has only shown better results on the first day in gelatin-free samples, but on the third day, it was better than the square scaffolds in all gelatin percentages. By the seventh day, scaffolds with 0% and 1% gelatin could not preserve albumin production better than the control group, and in 2% gelatin samples, the square scaffold had a slightly higher concentration. Based on previous data from the adhesion and viability of gelatin-free scaffolds, it can be concluded that this system is more effective for releasing albumin in the first few days, and adhesion gets better in the following days. As a result, it is reasonable to suppose that some of the cells in this medium have lost function over time, which is the opposite of the case with 2% gelatin scaffolds.

In the case of 2H and 2S, it has been observed that the system has a low attachment at first with good cell function, followed by reduced attached cells but a better function than other samples. This may result from gelatin degradation, washing off attached cells but presenting a new surface for the remaining cells to attach and

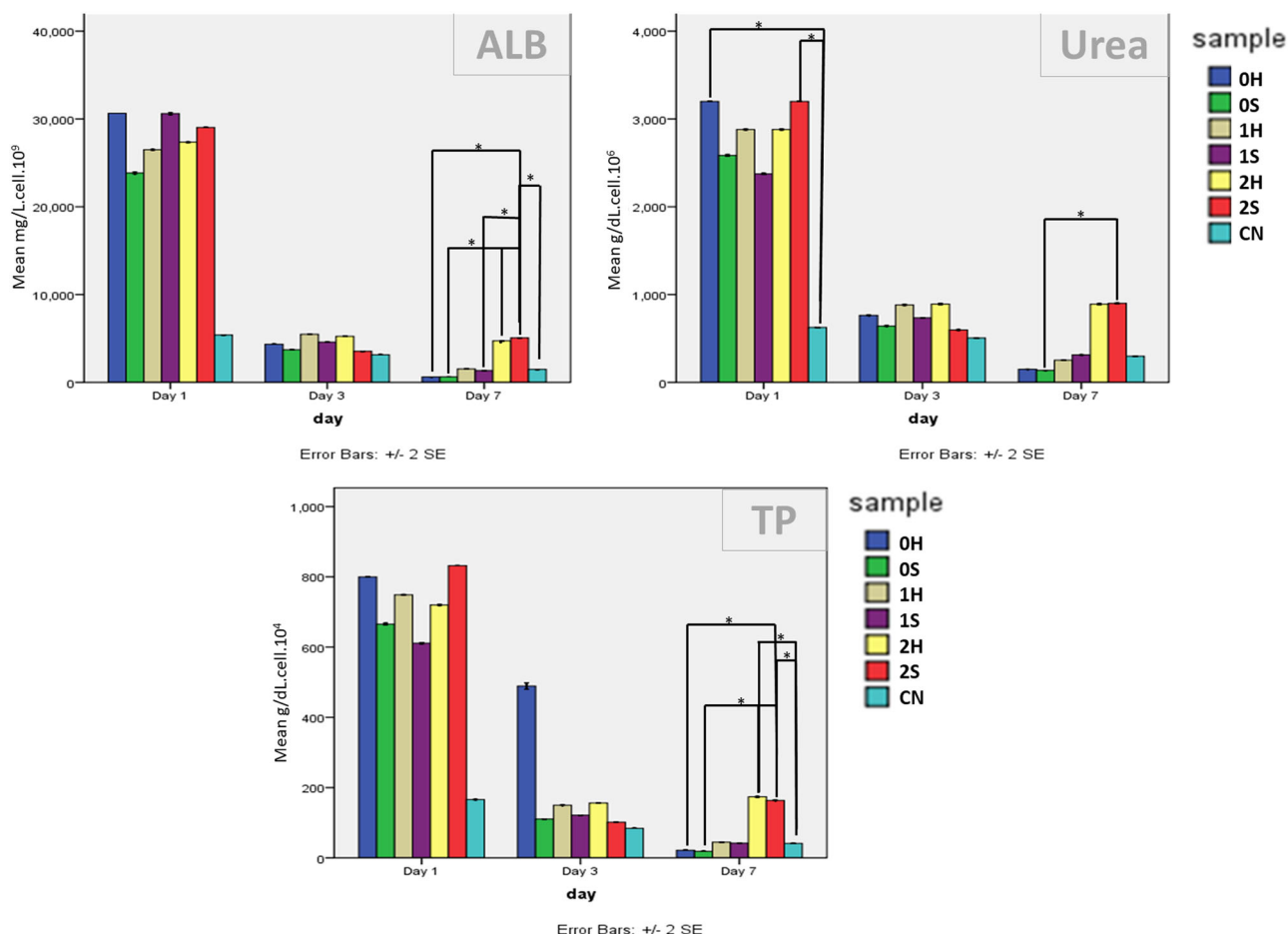


FIGURE 11 Liver cell functions: ALB, Urea, and TP of all samples in Days 1, 3, and 7. * show samples with p values below 0.05 in a single day. ALP, alkaline phosphatase; TP, total protein.

continue their function and proliferation. However, as concluded from the adhesion results, 2% gelatin coating reduces the effect of geometry.

Albumin production has been vastly assessed in LTE as the main indicator of liver-specific function. As you can see, the maximum albumin production in our study for the 0H sample on Day 1 is 30.64×10^{-6} mg/L cell and for the 2S sample on Day 7 is 4.056×10^{-6} mg/L cell. Compared with other studies, Ghaedi et al. (2012) could reach 0.5×10^{-6} mg/L cell albumin on Day 7 with culturing human bone marrow-derived mesenchymal stem cells (hMSCs) on a PLLA nanofiber scaffold. Also, Lee et al. (2017) observed a maximum of 0.01 mg/L cell of albumin within their 3D cell printed PCL construct and collagen bioink containing primary rat hepatocytes on Day 1 with a descending secretion rate to 0.005 mg/L cell on Day 7. In another study by Jia et al. (2020), the maximum amount of 1.3×10^{-9} mg/L cell albumin was measured in a microchip system with coculturing human hepatocyte (C3A) and human endothelial cells (EA.hy926) for 7 days.

Another fundamental function of the liver is urea secretion, in which the excess nitrogen in the body is metabolized. Otherwise, it will accumulate as toxic material causing brain and vascular damage (Leonard & McKiernan, 2004). As shown in Figure 11,

urea production of the hexagonal design on the first and third days was higher in the same gelatin concentration scaffolds except for 1S on Day 1. Regarding the scaffold design, cell function characterization has shown that during the first days of culture, the hexagonal scaffolds with 0% and 1% gelatin had better results. According to the adhesion data, we can understand the illuminating effect of scaffold architecture in preserving liver function despite lower attached cells.

Total protein (TP) comprises albumin and globular proteins (Alberghina et al., 2011). However, we cannot thoroughly compare TP results with microalbumin data due to differences in measurement tolerance. Yet, we can relate to them, respectively. The secretion of these biomolecules can show that the hepatocyte's phenotype is preserved. However, it would not guarantee the homogeneity of the engineered tissue (Lewis & Shah, 2016). According to Figure 11, TP secretion of 0H and 2H was the highest on the first day, indicating the hexagonal design to be more suitable in these systems than the square scaffolds. However, this superiority was more durable in scaffolds with 2% gelatin. On Day 3, all hexagonal scaffolds had higher secretion than their respective square design. This trend changed by Day 7, demonstrating 2% gelatin-coating to preserve TP production more efficiently and higher than the control group.

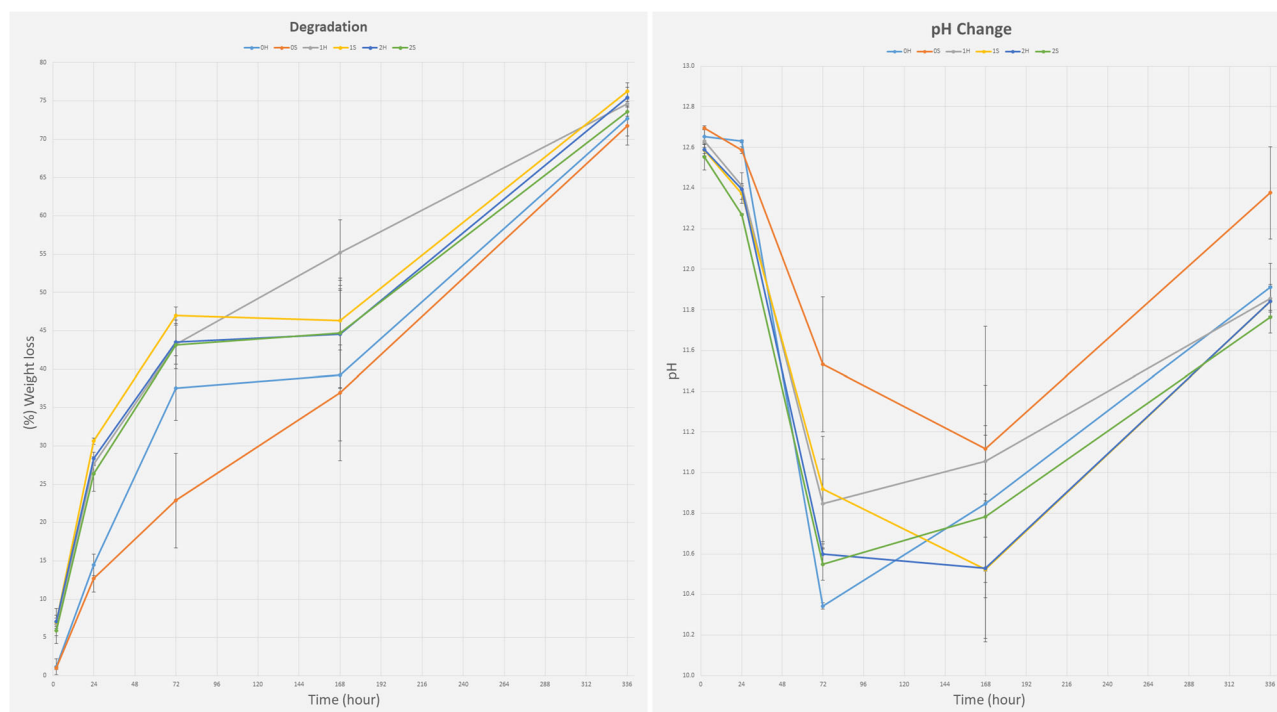


FIGURE 12 Degradation measured by decreased weight% and pH of all samples over time.

3.6 | Accelerated degradation

In this section, the weight loss of all samples and the pH of the environment are measurements for degradation in 2 h and 1, 3, 7, and 14 days as shown in Figure 12.

PLA degradation has been reported to start with hydrolysis, or the breakdown of ester bonds and products of low molecular weight acids decrease the pH (Li et al., 2004). Due to the lack of enzymes, gelatin degradation was a hydrolysis reaction. The breakdown of gelatinous amino acid chains can alter the environment's pH. Since gelatin degradation is very rapid, its effect will be seen early after exposure to the degradation solution (Tondera et al., 2016).

As shown in Figure 12, the highest degradation was for the 1S sample on Day 7, with 76.27% weight loss. The degradation process of all samples except for 0S and 1H was similar. It was very rapid in the first 72 h, resulting in a significant change in the pH and dissolution of the polymers. It can be deduced that the degradation occurred at the surface at this time, and the gelatin at the surface had begun to degrade. The samples treated by the Entrapment method were initially more susceptible to surface hydrolysis because of the initial hydrolysis step in this method. It can be assumed that the polymer chains on the scaffold surface have been partially hydrolyzed and continued to be hydrolyzed after immersion in NaOH solution, creating low-molecular-weight, acidic chains that reduced the pH.

The degradation rate of PLA in alkaline environments is generally a first-order function (Baltazar et al., 2020). After that, no significant weight loss was observed until Day 7 (168 h), resulting in a modest increase in pH. This can be attributed to the degradation and hydrolysis of the material bulk. This means that the pre-hydrolyzed

surface of the scaffold is degraded and that the bulk hydrolysis of the material has not reached a state to leak low-molecular-weight acidic chains. Also, on the seventh day, the scaffolds were so demolished that the junction areas of the struts, which were physically attached in the printing process by heat, had detached, and some fragments were floating in the degradation solution. This phenomenon also could indicate that in this period (from 3 to 7 days), the physical bonds and junctions have been disassembled, and fewer polymer chains have broken. Then, the weight loss and destruction rate increased by the 14th day (336 h). Despite the rapid degradation, the pH increased, which may be because, on the seventh day, the medium was replaced, and a new NaOH solution was added, leading to an increased amount of OH^- ions and partially neutralizing the acidic products. These weight loss data indicate that the degradation in our scaffold depends on the composition and surface chemistry due to the design of the two scaffolds that are arranged to have a roughly identical surface-to-volume ratio (about 10 mm^{-1}). It is expected that the degradation in the two scaffolds with the same surface chemistry and bulk will be close together.

4 | CONCLUSIONS

PLA scaffolds were successfully printed according to the intended designs with hexagonal and square cross sections. The gelatin coating was applied by a modified Entrapment method resulting in a more hydrophilic and rougher surface due to the partial surface hydrolysis. The tendency of liver cells to aggregate has resulted in more cell attachment on scaffolds without gelatin which grows over time.

Squared cross-sectional samples showed higher cell attachment than the hexagonal ones except in samples with 1% gelatin. However, the adhesion results are in contrast with the viability and cytotoxicity. These results showed the effect of geometry on these aspects. Albumin, urea, and TP production determined scaffolds with 2% gelatin could preserve more function and secrete more despite low cell attachment. However, all the samples had higher TP, urea, and albumin production the whole time than the control sample. We have successfully fabricated an LTE scaffold using FDM technology as an accessible fabrication method combined with feasible processing. It can be concluded that optimizing this system can deliver a functional liver scaffold to preserve cell activities for further studies. These data can help broaden the LTE field to fulfill the goals of modern biotechnology.

AUTHOR CONTRIBUTIONS

Elnaz Sadat Mirdamadi and Mehran Solati-Hashjin conceptualized the idea. Elnaz Sadat Mirdamadi, Zahra Khosrowpour, and Davod Javafri investigated experiments. Mehran Solati-Hashjin and Mazaher Gholipourmalekabadi had the role of project administrators and resources and were supervisors of this project. Mehran Solati-Hashjin was involved in funding acquisition for this research. Elnaz Sadat Mirdamadi took the lead in writing the manuscript, and Mazaher Gholipourmalekabadi with Mehran Solati-Hashjin reviewed and edited the draft of the manuscript. All authors provided critical feedback and helped shape the research, analysis, and manuscript.

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CONFLICT OF INTEREST

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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